

Deep Learning: Advanced Models and Methods (DL26)

Track 3: Graph-based Metric Learning for Scene Understanding

Gruppo: G18

Membri: [Salvatore Mario Carota]

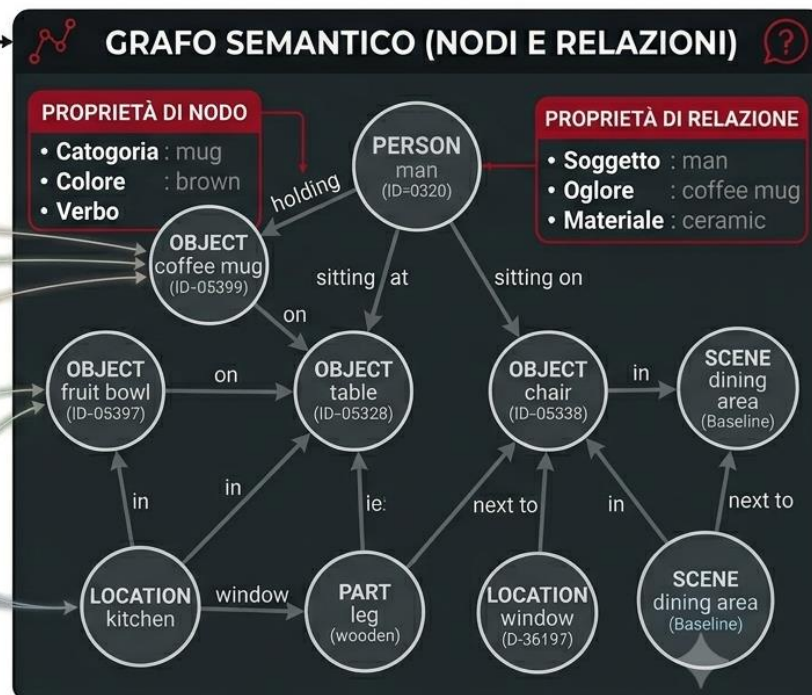
Repository:

<https://github.com/LenionK/dl26-projects>

From Pixels to Graphs: Representing Scenes as Structured Data



IL GRAFO CATTURA LE RELAZIONI
SEMANTICHE E SPAZIALI TRA GLI OGGETTI



OTTIMIZZATO PER RETRIEVAL E VQA



GQA: Visual Reasoning in the Real World

2328363



2379201



2387152



2379505



2407089



2330388



2334735



2327633



713232

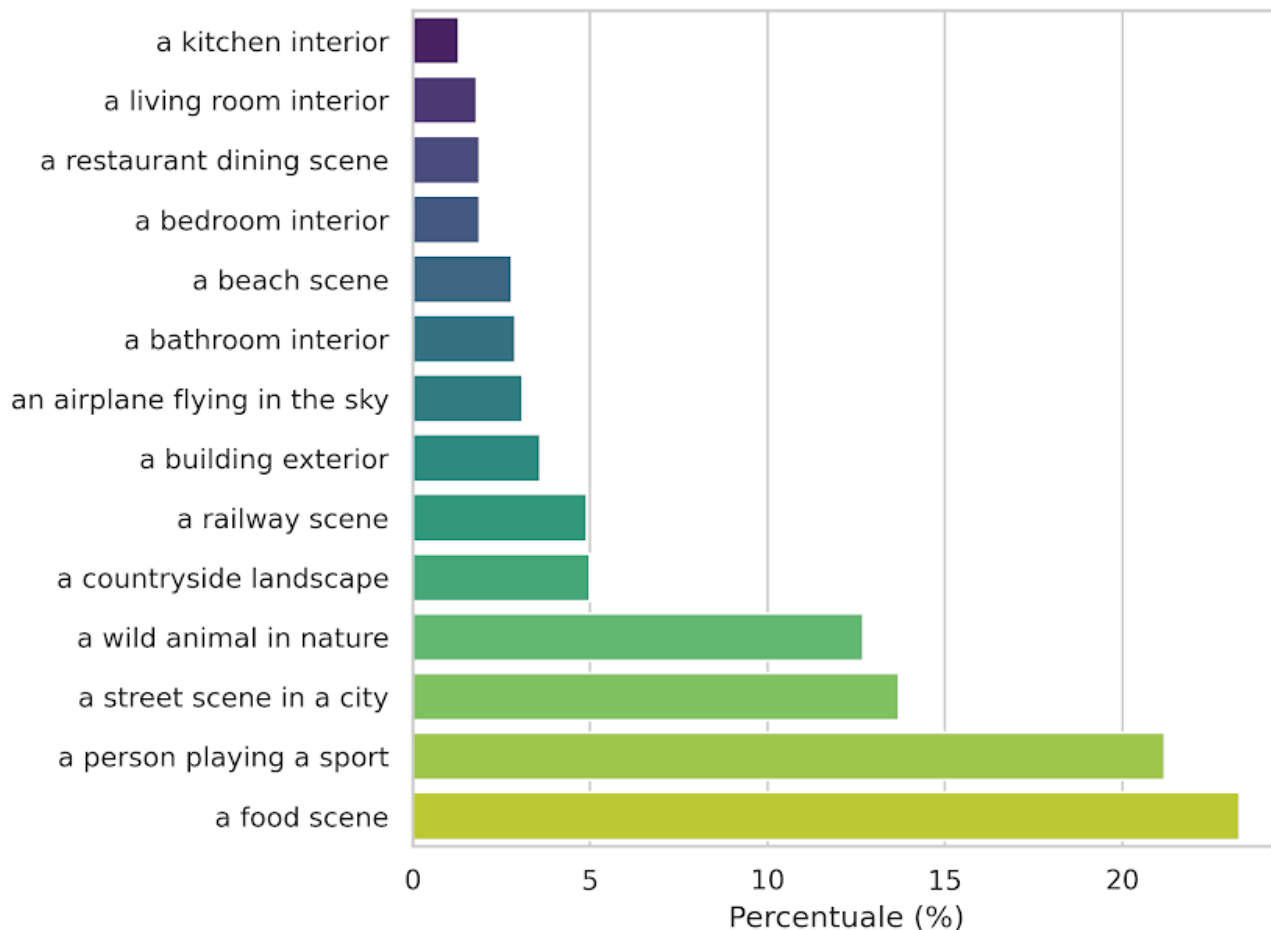


2414195



Bridging the Gap: Automatic Labeling via CLIP

Classi di Partenza
Interni
Urbano
Natura
Cibo



Tre encoder, stesso obiettivo metrico

Baseline - Graph - Contrastive Training

Modello	Input	Architettura	Embedding
Baseline	Immagine	ResNet-18 → FC	128-d
GCN	Grafo	3× GCNConv + pool mean/max	128-d
GINE	Grafo + edge type	4× GINEConv + pool mean/max	128-d

Supervised Contrastive Loss

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \frac{1}{|P(i)|} \sum_{p \in P(i)} \log\left(\frac{\exp(z_i \cdot z_p / \tau)}{\sum_{j \neq i} \exp(z_j \cdot z_i / \tau)}\right)$$

Training e iperparametri

Parametro	Baseline	GCN/GINE
Optimizer	Adam ($lr=10^{-4}$)	Adam ($lr=3e-4$, $weight_decay=1e-4$)
Scheduler	-	CosineAnnealingLR
Grad clipping	-	$Max_norm=1.0$
Batch sampling	6 classi x 4 campioni	6 classi x 4 campioni
Epochs	10	20

Scene-To-Scene retrieval con metric learning

Retrieval Evaluation: Given a query scene graph, perform retrieval on the dataset to fetch structurally similar graphs, evaluating via standard classification metrics (Accuracy, Precision, Recall).

CNN

QUERY



#1 a food scene
0.984



#2 a food scene
0.984



#3 a food scene
0.984



GCN

#1 a food scene
0.998



#2 a food scene
0.997



#3 a food scene
0.997

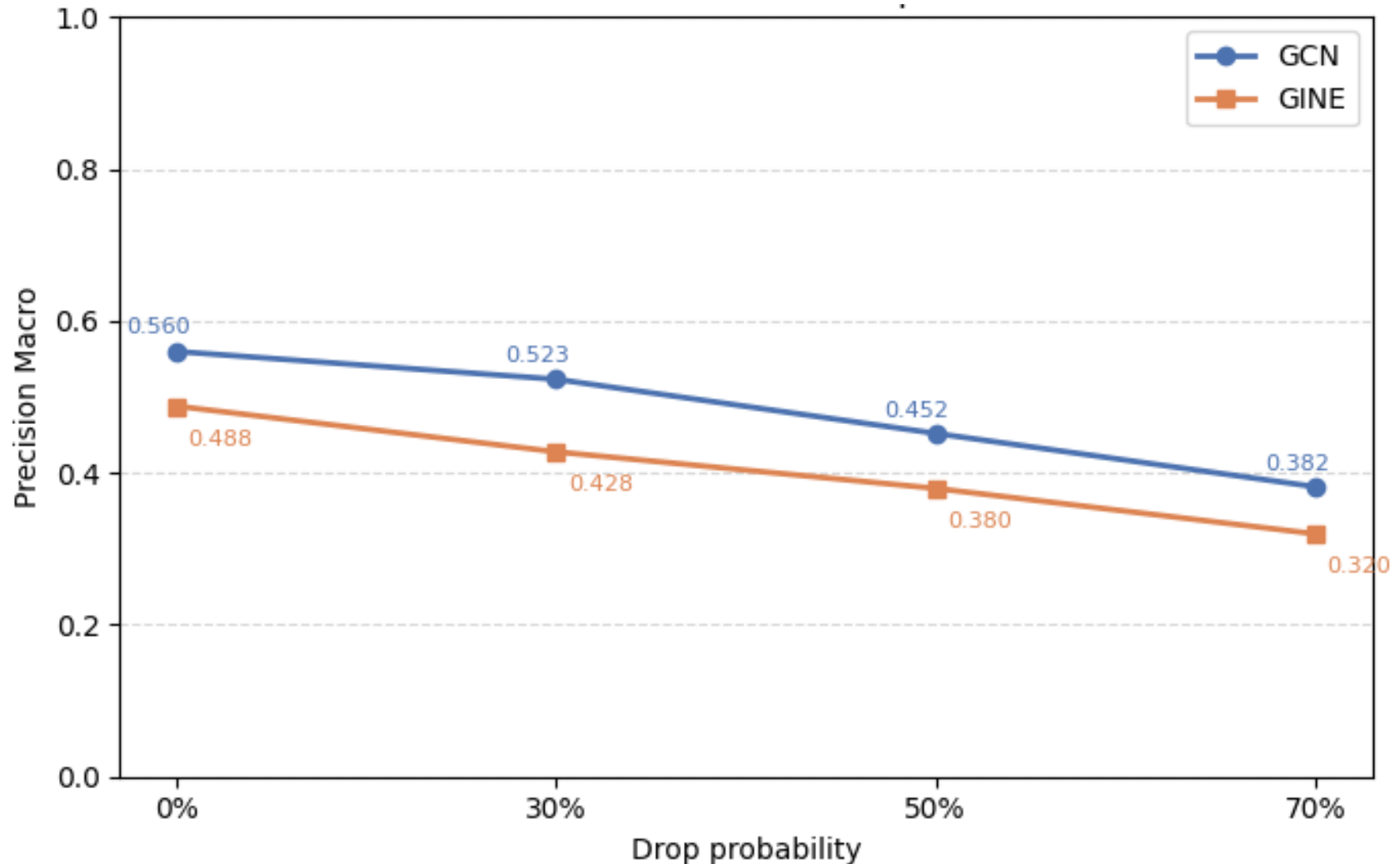


Metriche: Scene-To-Scene retrieval con metric learning

Metric	Baseline	GCN	GINE
acc@1	0.6231	0.5966	0.4991
precision_macro	0.5855	0.5673	0.4877
recall_macro	0.5668	0.5661	0.4892
recall@1	0.6231	0.5966	0.4991
recall@10	0.8881	0.8791	0.8572
precision@1	0.6231	0.5966	0.4991
precision@10	0.6152	0.6046	0.4978
mrr	0.7163	0.6949	621
map	0.68	0.6631	0.5778

GNN degrada gradualmente anche con il 70% dei nodi rimossi

Extra Objective: robustezza alle perturbazioni strutturali



GNN degrada gradualmente anche con il 70% dei nodi rimossi

Extra Objective: robustezza alle perturbazioni strutturali

drop Prob.	GCN			
	Precision (macro)	Recall (macro)	Acc@1	MAP
0%	0.5598	0.5538	0.5966	0.6594
30%	0.5232	0.5188	0.5592	0.6236
50%	0.4519	0.4567	0.5103	0.5821
70%	0.3817	0.3822	0.4514	0.5299

GRADCAM++

Extra Objective: Visualize and interpret which edges/relations act as critical focal points for the model's similarity metric.

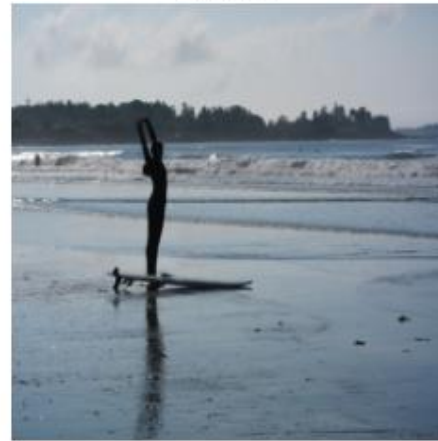
QUERY
a beach scene
originale



QUERY
a beach scene
GradCAM++



#1 MISS
a person playing a sport (0.98)
originale



#1 MISS
a person playing a sport (0.98)
GradCAM++



#3 MISS
a wild animal in nature (0.98)
originale



#3 MISS
a wild animal in nature (0.98)
GradCAM++



#4 OK
a beach scene (0.98)
originale



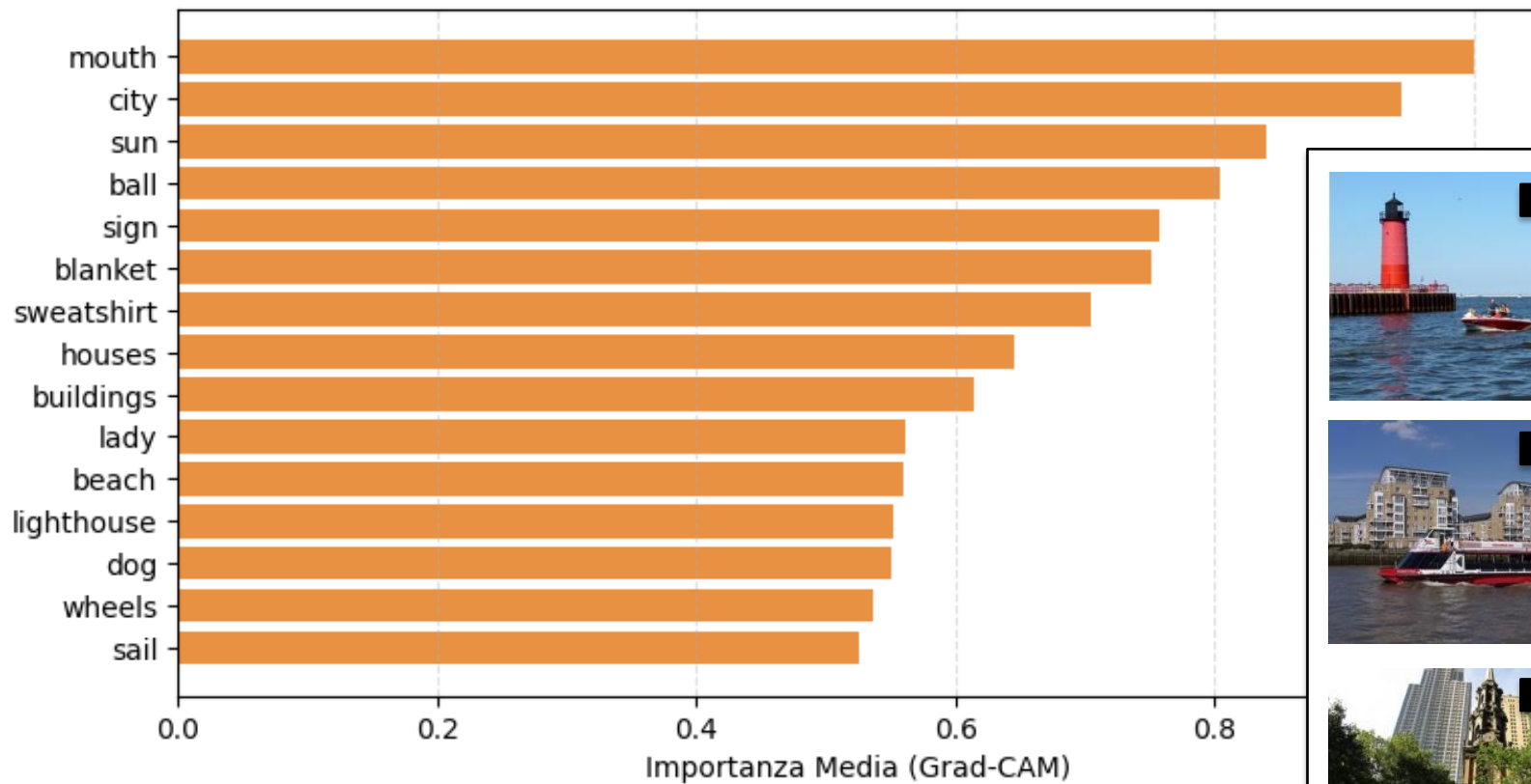
#4 OK
a beach scene (0.98)
GradCAM++



GRAPHGRADCAM

Extra Objective

GCN — Classe: a beach scene
(media su 100 campioni, top-15 nodi)



Focal Point Analysis — Ranking Disruption

Extra Objective: Visualize and interpret which edges/relations act as critical focal points for the model's similarity metric.

GCN

Relazioni	Disruption
leaves → tree	0.700000
sky → cloud	0.560000
cloud → sky	0.540000
tree → leaves	0.540000
clouds → sky	0.433333
sky → clouds	0.400000
water → boat	0.360000

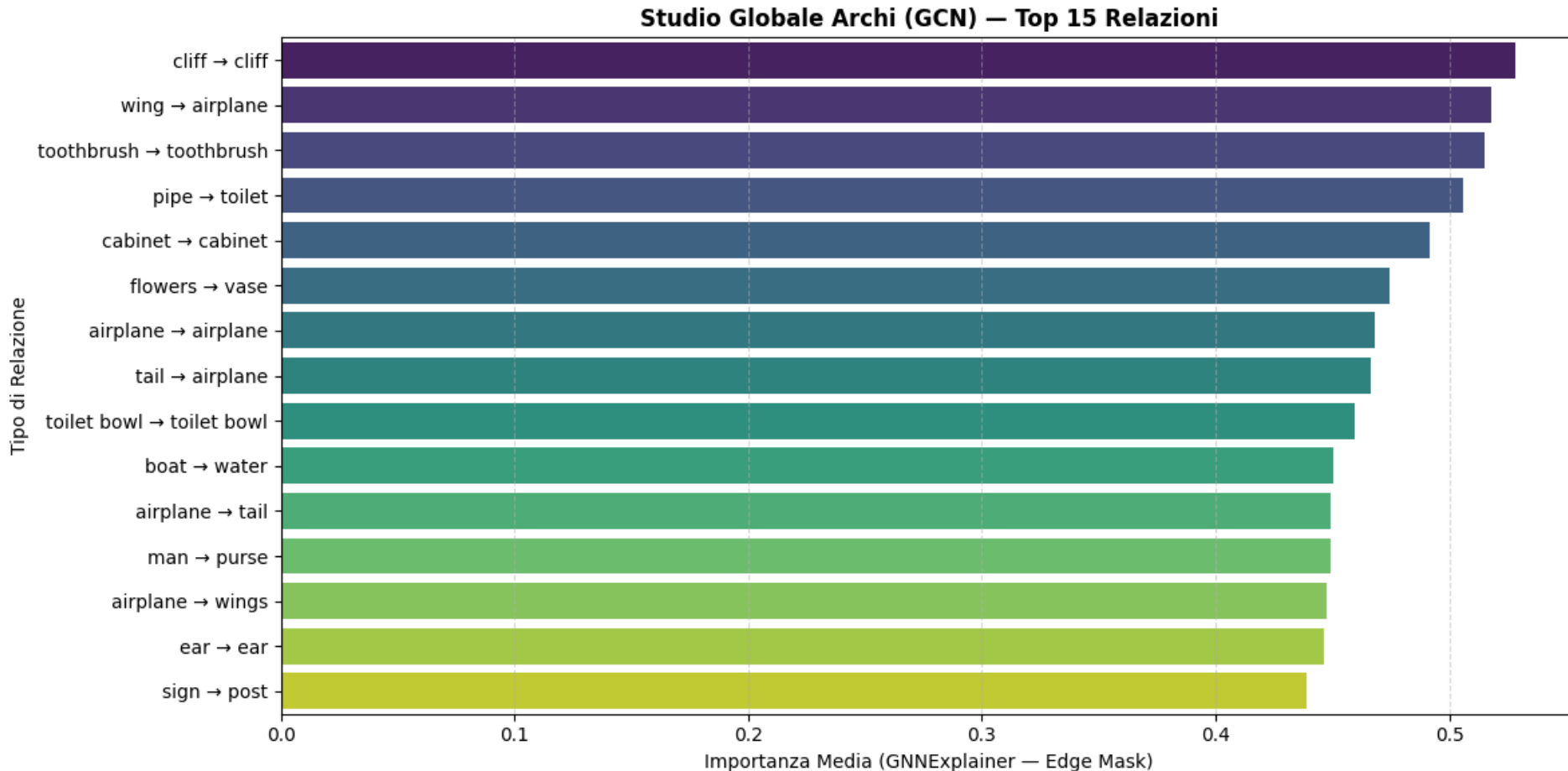
GINE

Relazioni	Disruption
to the left of	0.480000
to the right of	0.448148
on	0.218657
in	0.120896
wearing	0.094074
near	0.061481
of	0.059701

$$\text{Disruption} = 1 - \text{overlap}(\text{top-k base}, \text{top-k perturbed})$$
$$\text{overlap} = |\text{top-k}(\text{base}) \cap \text{top-k}(\text{perturbato})| / k$$

GCN: Studio Archi – GNNExplainer

Extra Objective: Visualize and interpret which edges/relations act as critical focal points for the model's similarity metric.



GINE: Studio Archi – GNNExplainer

Extra Objective: Visualize and interpret which edges/relations act as critical focal points for the model's similarity metric.

